

Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis

Daniela Denier¹, **Yiming (Emmett) Peng**^{2,3}, Kuan Liu², Amanda Ricciuto^{2,3}, on behalf of the Pediatric PSC Consortium

¹University of Bern, ²University of Toronto, ³The Hospital for Sick Children

Introduction

Background

Primary sclerosing cholangitis (PSC) is a chronic cholestatic liver disease with heterogeneous outcomes. While normal one-year gamma-glutamyl transferase (GGT) predicts better prognosis, longitudinal GGT patterns remain unclear.

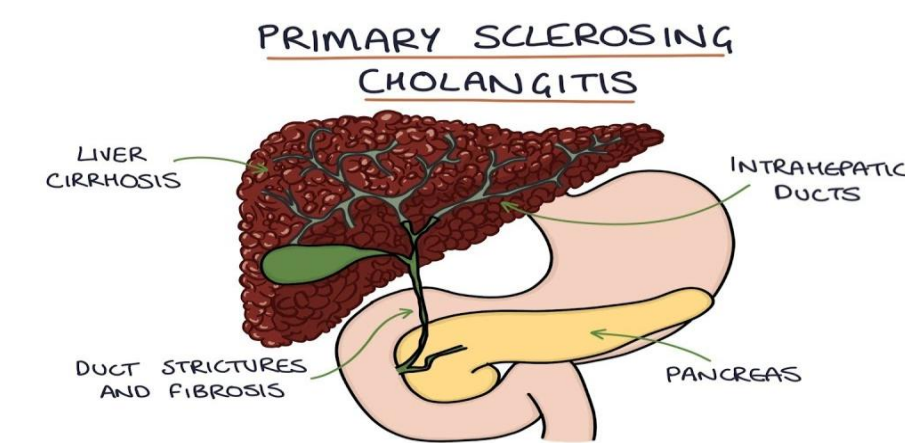
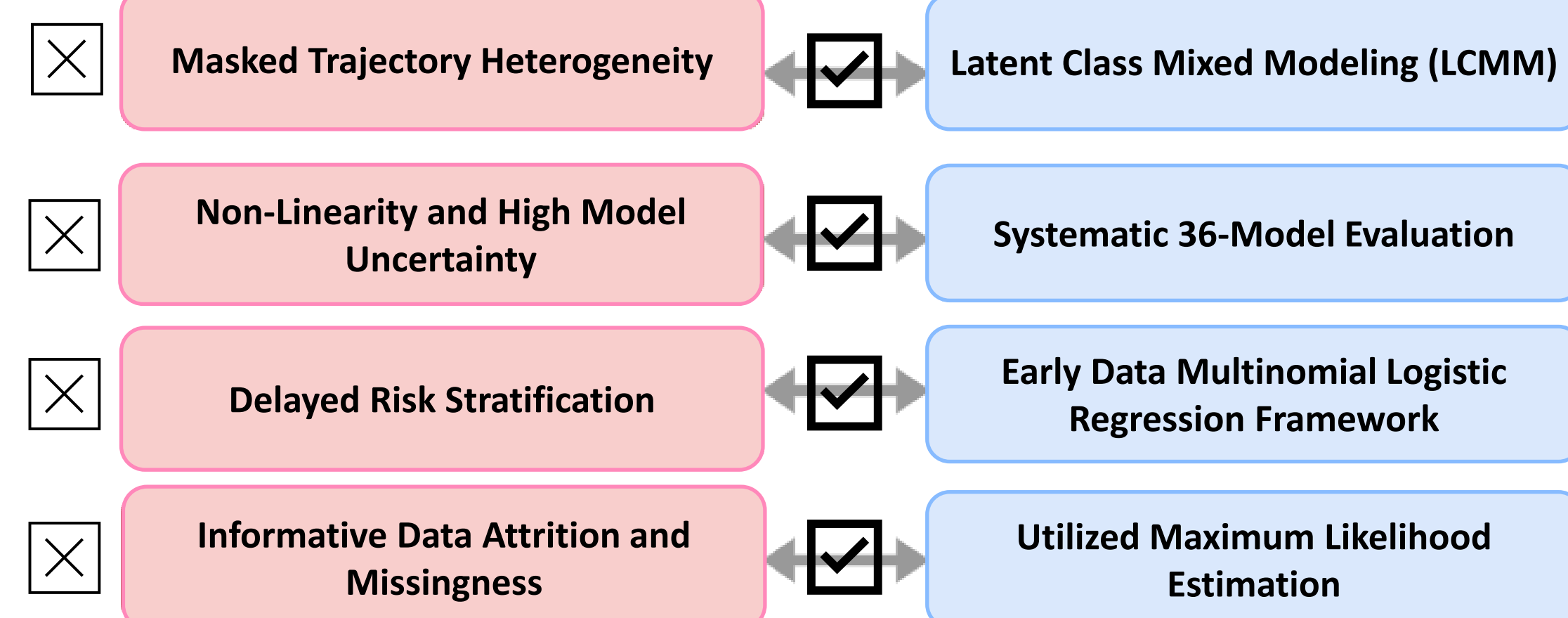


Image Credit: <https://www.youtube.com/watch?v=ZXs6FjnBIs>

Challenges and Innovation



Objectives

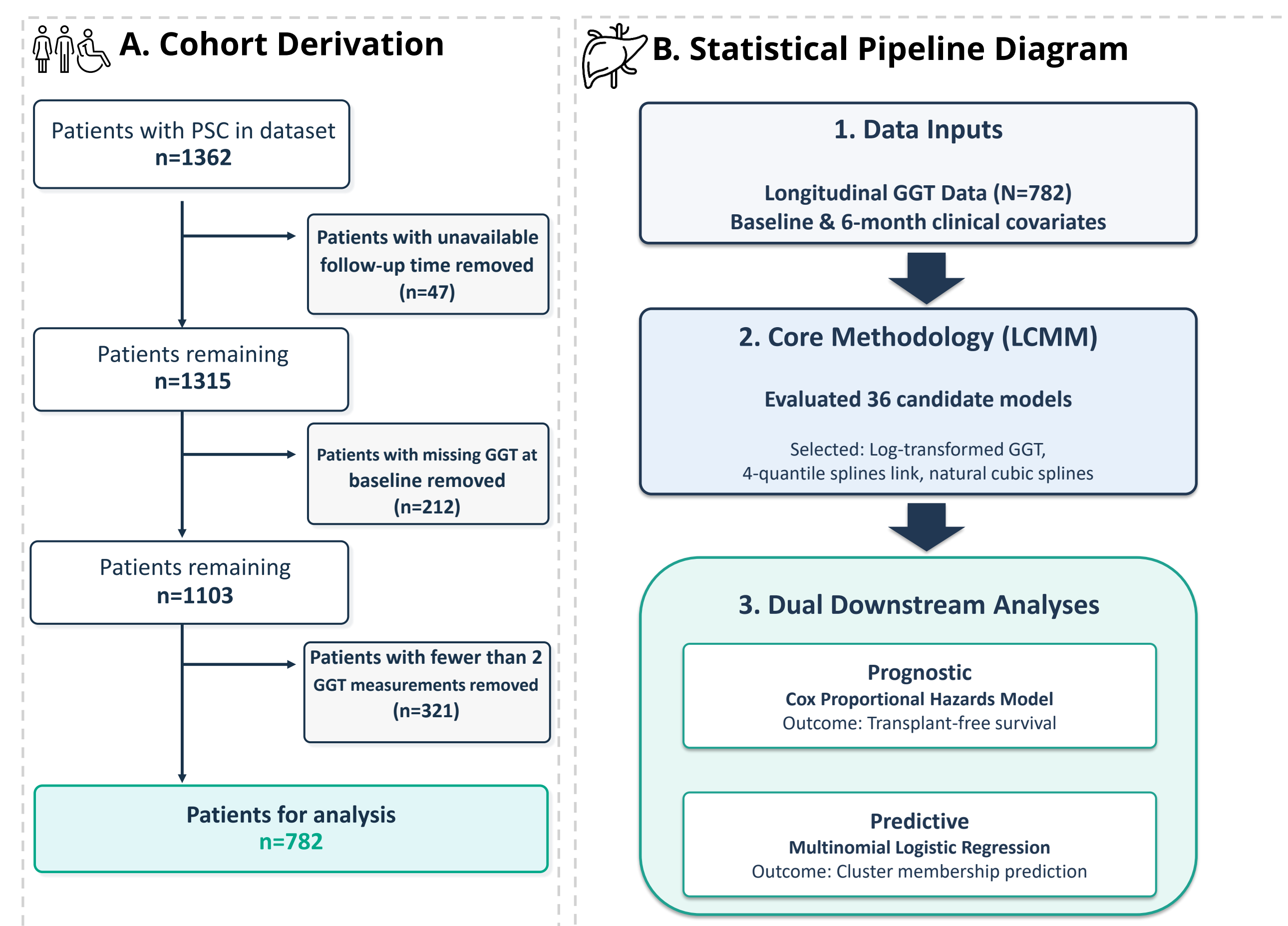
To apply **latent class mixed modeling** to longitudinal GGT data to uncover **distinct prognostic phenotypes** in pediatric PSC, evaluate their impact on **transplant-free survival**, and develop early prediction models for cluster membership.

Data

Study cohort

- Data Source:** 782 pediatric patients from the international Pediatric PSC Consortium.
- Inclusion Criteria:** Patients required a baseline GGT measurement and at least one follow-up GGT measurement within 3 years of PSC diagnosis.

Study Population & Analytical Pipeline (Figure 1)



Methodology

Developed an integrated **statistical pipeline** to identify unobserved **longitudinal biomarker trajectories**, evaluate their prognostic impact, and predict **cluster membership** early in the disease course:

Module 1 – Latent Class Mixed Modeling (LCMM): Identify distinct longitudinal GGT trajectory clusters from highly variable serial measurements.

- Formulation:** Let Y_{ij} denote the observed GGT measurement for patient i at time t_{ij} . Conditional on the latent class $g \in \{1, 2, 3, 4\}$, the model is defined as:

$$\Lambda(Y_{ij}; \eta) = X_{ij}\beta_g + Z_{ij}u_{ig} + \epsilon_{ij}$$

- Design Highlights:** $\Lambda(\cdot; \eta)$ uses a **4-quantile splines link function** to normalize log-transformed GGT data. The fixed (β_g) and random (u_{ig}) effects include natural cubic splines to capture non-linear temporal trends.
- Model Selection & Missing Data:** Evaluated 36 candidate models, selecting the optimal fit using **Bayesian Information Criterion (BIC)** and classification quality (**entropy > 0.6**). Missing longitudinal data were handled conditionally under the Missing at Random (MAR) assumption using **maximum likelihood estimation**.

Module 2 – Prognostic Modeling: Evaluate the association between the identified trajectory clusters and the primary clinical endpoint (transplant-free survival).

- Formulation:** Employed a Cox Proportional Hazards model where the hazard $h_i(t)$ for patient i is:

$$h_i(t) = h_0(t) \exp(\gamma^T W_i + \theta C_i)$$

- Design Highlights:** C_i represents the assigned **LCMM trajectory cluster**. W_i controls for baseline clinical covariates including age, sex, IBD status, AIH overlap, large duct involvement, and medication exposure.
- Validation:** The **proportional hazards assumption** was verified by examining the interaction between covariates and time based on scaled Schoenfeld residuals.

Module 3 – Predictive Modeling: Predict long-term trajectory cluster assignment early in the disease course for clinical risk stratification.

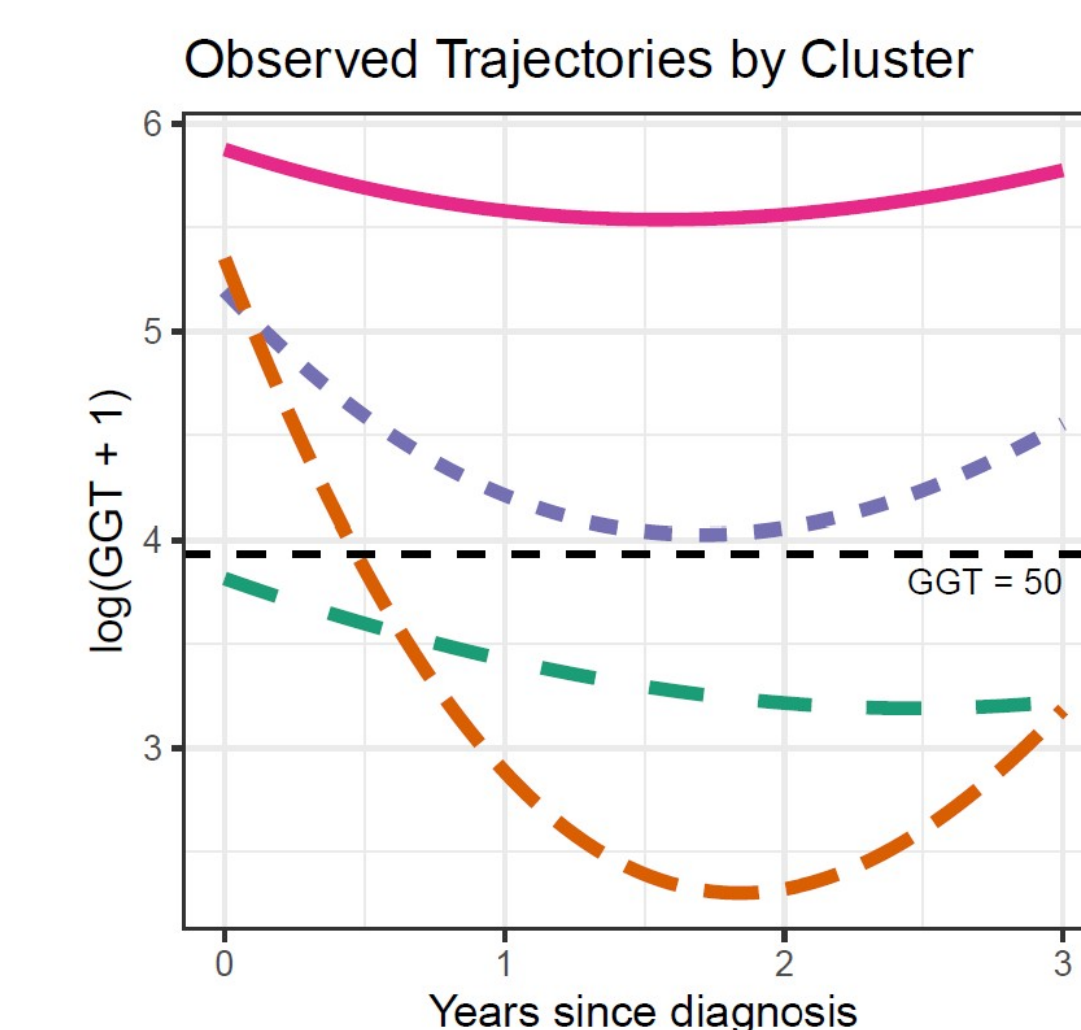
- Formulation:** Developed a multinomial logistic regression framework using only early disease markers. Let $C_i \in \{1, 2, 3, 4\}$, with Cluster 4 (Persistently Low) as the reference class:

$$\log(P(C_i = g) / P(C_i = 4)) = \alpha_g + \omega_{1g} \log(\text{GGT}_{\text{base},i}) + \omega_{2g} \log(\text{GGT}_{\text{6mo},i})$$

- Design Highlights:** The model prioritizes early clinical use by restricting inputs to **baseline and 6-month log-transformed GGT values**.
- Evaluation:** Data were randomly split into **70% training and 30% held-out test sets**. Predictive performance was evaluated on the test set using multiclass and per-class area under the receiver operating characteristic curve (AUC).

Results

- Figure 2:** LCMM identified four distinct GGT trajectory clusters: persistently elevated (23.1%), moderate-stable (29.8%), elevated-declining (26.0%), and persistently low (21.1%).
- The moderate-stable and elevated-declining clusters had similarly elevated baseline GGT but followed different longitudinal paths, showing that baseline GGT alone misses important prognostic information. (**Figure 2**)
- Figure 3:** Five-year transplant-free survival differed across clusters: 74%, 86%, 98%, and 96% for Clusters 1–4, respectively (log-rank $p < 0.0001$).
- Table 1:** Baseline and 6-month GGT values predicted long-term trajectory membership well (multiclass AUC = 0.835).



Model	Predictors	Multiclass AUC	Cluster 1 AUC	Cluster 2 AUC	Cluster 3 AUC	Cluster 4 AUC
1	GGT (baseline, 6m) log-transformed	0.835	0.834	0.768	0.860	0.859
2	GGT + demographic + clinical	0.838	0.831	0.760	0.878	0.865
3	GGT + demographic + clinical + all labs	0.813	0.831	0.699	0.852	0.850
4	GGT + ALT (baseline, 6m)	0.831	0.829	0.751	0.869	0.856

Figure 2: Identification of Latent Trajectory Phenotypes

Table 1: Early Prediction of Long-Term Trajectories

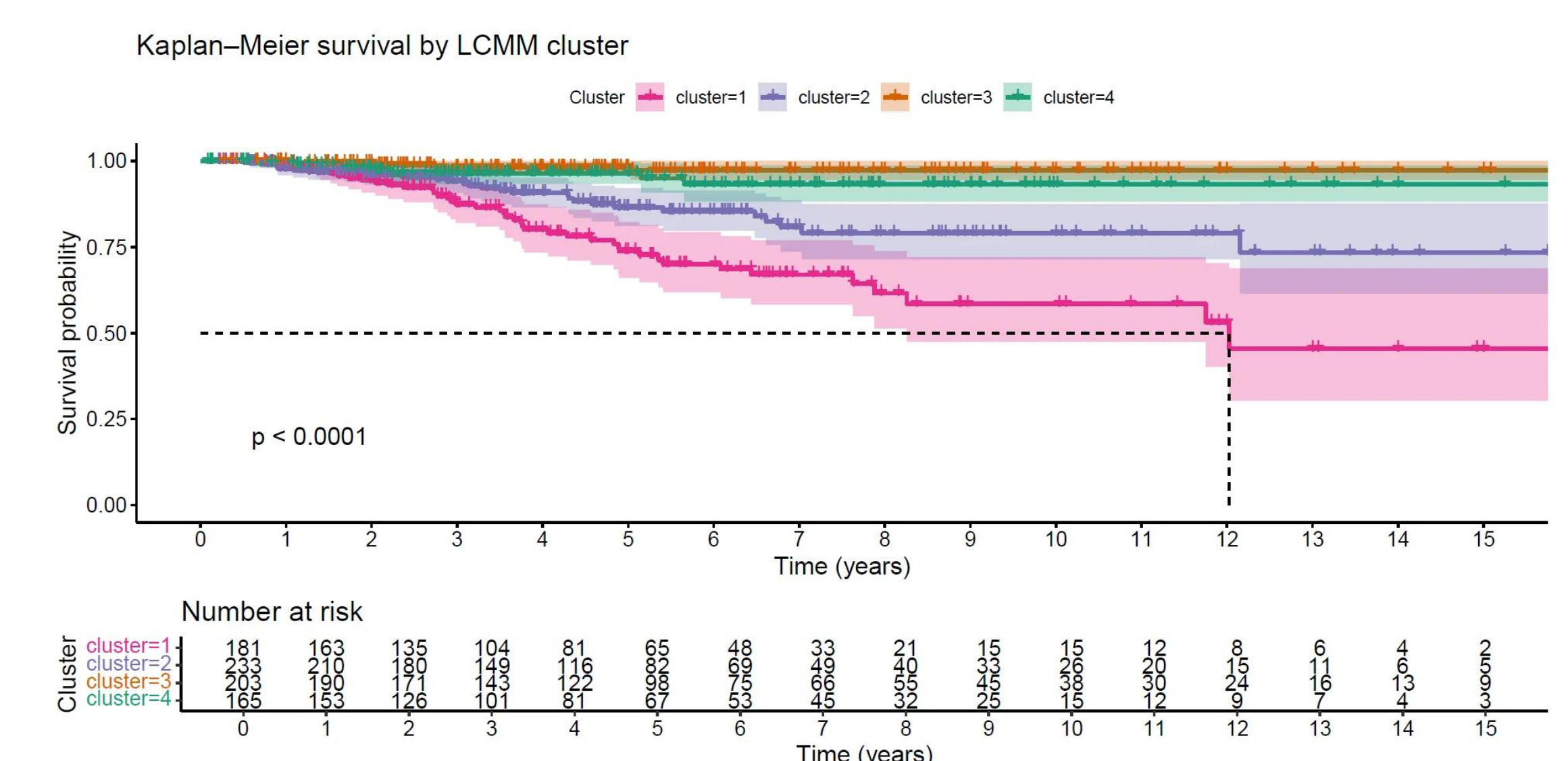


Figure 3: Prognostic Impact on Transplant-Free Survival

Conclusion

- Longitudinal GGT trajectory modeling identified four clinically meaningful PSC phenotypes with distinct prognoses.
 - Early GGT response, rather than baseline GGT alone, helped distinguish patients with different long-term outcomes.
 - These findings support trajectory-based risk stratification and may inform future PSC trial design.
- Limitations and Future work**
- Limitations include retrospective design, possible selection bias, routine-care measurement timing, and possible treatment effects.
 - Future work should validate these trajectory phenotypes in independent prospective cohorts.